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Q1

Binary search is $O(n)$ in the worst case because if the element is missing you may have to look at every cell to be sure. The recurrence just shows the best case is faster.

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Q2

Exchange argument: let $G = g_1..g_k$ be greedy's picks (by finish time) and $OPT = o_1..o_m$ an optimal solution, both sorted. By induction $f(g_i) \leq f(o_i)$: greedy picks the earliest-finishing compatible interval, so we can swap o_i for g_i without breaking compatibility. Hence $k \geq m$, so greedy is optimal.

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Q3

DP, $O(n^2)$.

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Q4

True. Invariant: when BFS finishes layer d , every vertex at distance d has been discovered with $\text{dist} = d$. Base: layer 0 is $\{s\}$. Step: any v at distance $d+1$ has a neighbour u at distance d ; when u is dequeued v is discovered with $\text{dist } d+1$, and no earlier layer can reach v or its distance would be $< d+1$.